MMN realness checks at FCz — N-effect and deviance scaling

aux/analysis_MMN_realness_checks/mmn_dose_response_fcz_memo.md · rendered from markdown

Contents

MMN "realness" at FCz — the N-effect and deviance scaling of the S7 trough

Two dose-response checks on whether the in-silico MMN behaves like a human/clinical MMN. Both ask the same question — **does the trough deepen as the paradigm makes the oddball more surprising?** — along two axes the MMN literature has strong priors about:

- **The N-effect.** More standards between deviants → deeper MMN.
- **Deviance scaling.** Larger standard→deviant frequency separation → deeper MMN.

Site: the FCz electrode only, mTRF only, six models (whisper-large excluded — see §5.6). **Gate:** S7@0.75 is the headline, with a fully parallel S2 (shape-only) companion. **Status:** complete. The per-trial re-run landed 2026-08-10; all three verification gates passed, including a byte-level reproduction of the committed FCz S2/S7 counts.

Figures are referenced by path relative to this file. `README.md` in this directory is the figure-reading guide and should be read first if the panels are unfamiliar.

The result in three sentences

Both manipulations increase the model's MMN amplitude, but the effects are small and the gated amplitude figures hide them. Conditioning on a −0.75 μV floor truncates exactly the traces a weak dose-response moves, so the S7-gated trough looks flat while the *probability of clearing the floor given an MMN-shaped response* rises sharply with both N ($z = +6.63$, $p = 3e-11$) and deviance ($\rho = +0.071$, $p = 0.015$). The honest summary is not "occurrence responds but magnitude doesn't" — it is that **amplitude responds weakly, and every censored view of it understates the effect** (§3).

---

1. Design, in brief

Two condition sources, reported side by side and never pooled into a single regression:

	LIT	NOVEL-P2
conditions	48 = 24 literature Frequency methods ×	254 = 127 selected pairs ×
semitone range	0.84 – 12.0 (10 values)	4.50 – 63.0 (122 values)
timing	SOA 200–1000 ms, duration 50–200 ms — **varies**	580 ms / 80 ms / 10% — **fixed**
selection	none	**selected on the outcome** (≥5/6 model agreement in phase 1)
S7@0.75 at FCz	125 / 288 rows	856 / 1524 rows

Three analysis sets follow: `lit` (48 conditions), `lit_p2` (302), `p2` (254). A fourth `p2_top100` panel appears on the pooled figures only and is described in §5.7.

Two gates. `S7@0.75` = an interior trough in 100–240 ms recovering ≥50% within 120 ms (**S2**), *and* trough ≤ −0.75 μV. `S2` alone drops the depth requirement. The floor is not a minor filter: it discards **52% of LIT's S2 traces and 28% of NOVEL-P2's**. Every S7 amplitude number in this memo is therefore a **lower bound**.

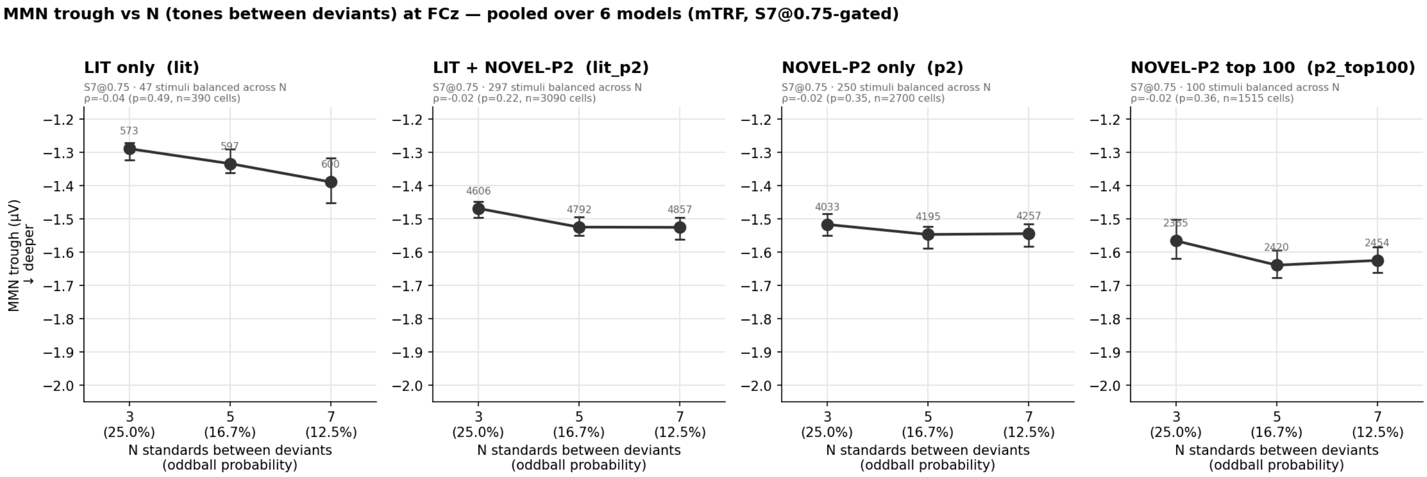
One summary statistic in every panel: the median with a 95% bootstrap CI of the median. The per-bin trough distributions are strongly skewed (0 of 31 consistent with normality, Shapiro–Wilk; skewness −2 to −3), so a mean is dragged toward the tail. `__mean_sem` companions of every centre-and-interval figure are provided on identical axes so the choice can be audited; the N=3→7 change agrees between the two statistics to ≤0.02 μV, so only the level moves, not the trend.

2. The N-effect

Per-trial scoring of 27,180 deviants (6 models × 302 conditions × 15 trials). The path reproduces the pipeline's own stored `n7v1_peak` to 2.1–3.0 × 10^{-7}, so this is the committed criterion applied to single trials, not a re-implementation.

All N amplitude figures use a *balanced* set — only stimuli producing a criterion-passing MMN at N=3 *and* 5 *and* 7. Without it the gate admits a different population at each N and the mean moves because membership changed rather than because anything deepened. That was not hypothetical: the unbalanced LIT panel rose visibly (unpaired p = 0.003) while paired within condition it was null (p = 0.23, 50% of stimuli — a coin flip).

2.1 Gated amplitude: flat



Median with 95% bootstrap CI over the S7@0.75-passing trials of all 6 models; grey numbers = trials per point.
BALANCED SET: only stimuli that pass the criterion at N=3 AND 5 AND 7 are included, so N is a within-stimulus manipulation and a change across N cannot be a change in which stimuli are averaged. Shared y-axis across the three sets. ρ is Spearman on one value per (model, condition, N) cell, NOT on raw trials — the 5 variations of a cell are the same paradigm re-rolled and are not independent. N is confounded with oddball probability by construction: the generator sets rare-tone probability to 1/(N+1), so N = 3/5/7 means 25%/16.7%/12.5% (shown on the x-axis). A trough that deepens across N is MMN-like but cannot be attributed to local spacing rather than global rarity.
The 4th panel is the top 100 of the search's own phase-2 ranking (by model agreement, then trough depth) — SELECTED ON THE OUTCOME on top of NOVEL-P2's own selection, so its troughs are deeper by construction. It bounds the best responders; it does not estimate an effect size.
Balancing removes the composition confound that made the earlier unpaired version of this figure misleading.
n_effect_change_table.csv gives the average within-stimulus change in μV from N=3→5→7, and n_effect_paired_n3_vs_n7.csv the paired significance test.

Spearman ρ of trough vs N, computed at the (model, condition, N) cell level — the 5 variations within a cell are the same paradigm re-rolled and are not independent, so a raw-trial ρ would inflate n ~5×:

set	ρ (S7)	p	ρ (S2)	p
`lit`	–0.064	0.185	–0.023	0.513
`lit_p2`	–0.015	0.378	**–0.039**	**0.0079**
`p2`	–0.006	0.754	**–0.041**	**0.0117**

Gated, nothing is significant. **Ungated, both NOVEL-P2-containing sets are.** That pattern — an effect that appears when the floor is removed — recurs throughout this memo and is the subject of §3.

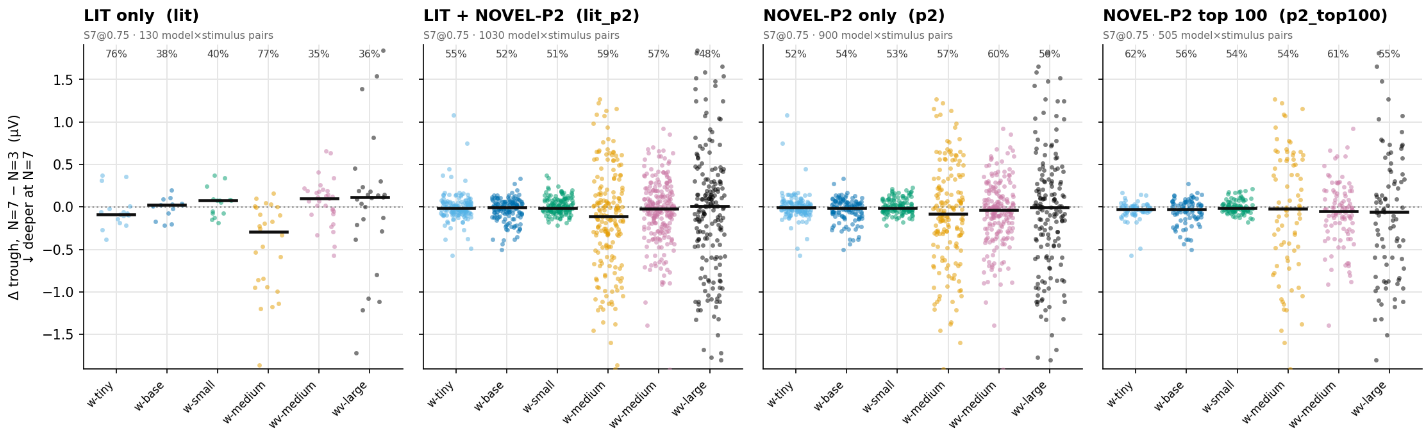
The paired within-condition test agrees and quantifies the gap:

set	Δ (N7 – N3), S7	p	Δ, S2	p
`lit`	–0.065 μV	0.228	–0.054 μV	0.116
`lit_p2`	–0.049 μV	0.0021	**–0.080 μV**	**2.2e-05**
`p2`	–0.047 μV	0.0041	**–0.085 μV**	**8.7e-05**

The floor hides roughly half the amplitude effect. –0.085 μV ungated against –0.047 μV gated.

2.2 The raw-data view: change is near-symmetric about zero

Within-stimulus change in MMN trough across N at FCz — (S7@0.75-gated)



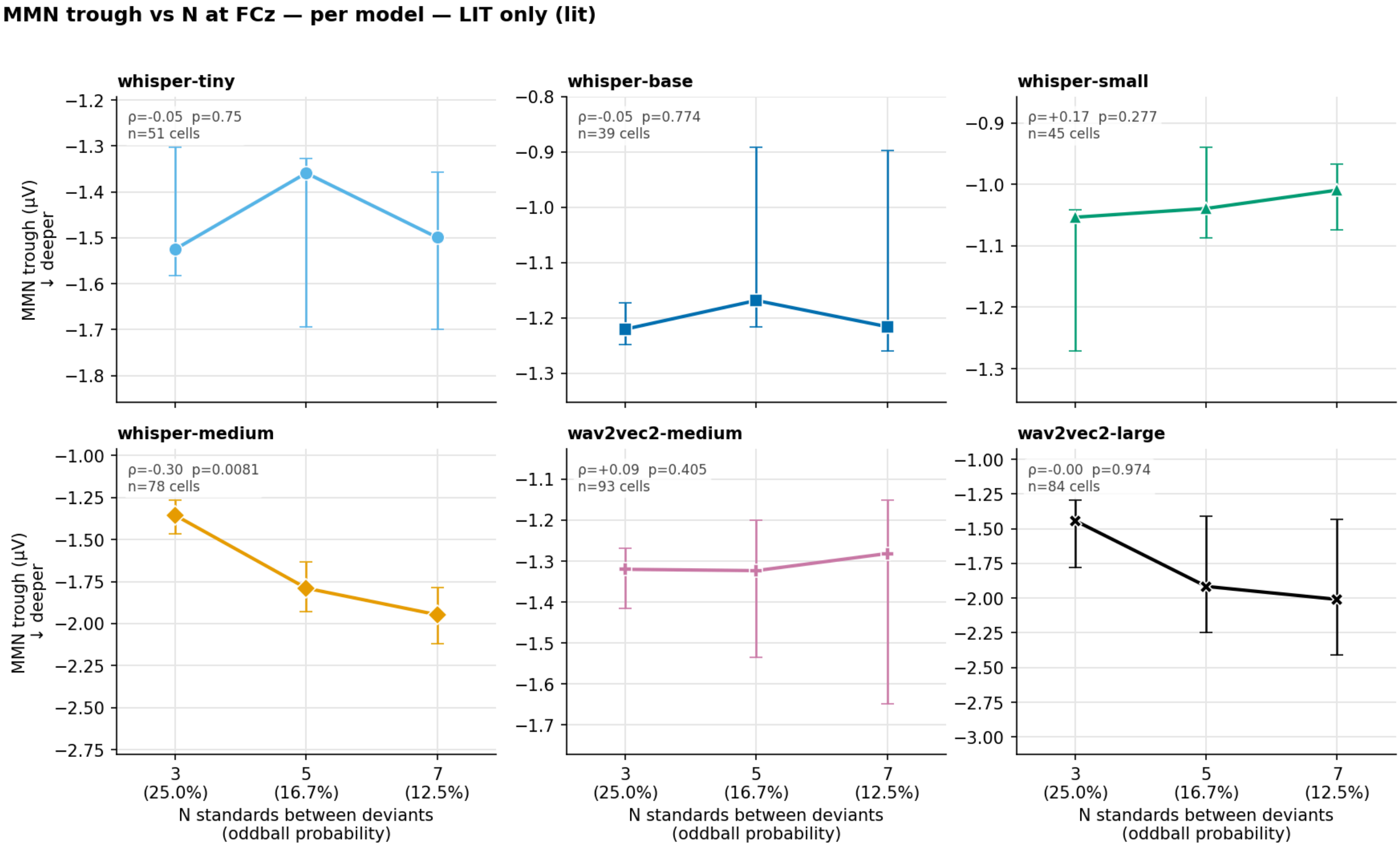
One point per (model, stimulus): that stimulus's own trough at N=7 minus its trough at N=3, so each point is a within-stimulus change and the comparison needs no baseline correction. Black bar = median; the % above each column is the share of stimuli that deepened. The axis is symmetric about zero so neither direction is visually favoured, and clipped at the 99th percentile of |Δ|.
This replaces a raw-trial scatter (the deviance figures' treatment) for two reasons: N has only 3 x-levels carrying 4,000+ trials each, so a strip plot is a solid block in which the ~0.05 μV effect is invisible; and trials within a stimulus are not independent, so a trial scatter would imply ~14,000 independent observations. What it shows that an error bar cannot: the change distribution is roughly ±1 μV wide and nearly symmetric, so a 54–55% deepening share is barely better than a coin flip.

This is the N analogue of the deviance scatter, but plots the **within-stimulus change** rather than raw trials — N has only 3 x-levels carrying 4,000+ trials each, so a strip plot is a solid block, and trials within a stimulus are not independent, so a trial scatter would imply ~14,000 independent observations.

set	n (model × stimulus)	% deepening	median Δ	IQR of Δ
`lit`	130	50%	−0.002 μV	−0.20 … +0.14
`lit_p2`	1030	54%	−0.019 μV	−0.17 … +0.11
`p2`	900	55%	−0.020 μV	−0.16 … +0.11
`p2_top100`	505	57%	−0.026 μV	−0.17 … +0.09

At the level of individual stimuli the N effect is barely distinguishable from a coin flip. The change distribution is ~±1 μV wide against a median shift of 0.02 μV. This is true, important, and invisible in any figure that plots only the centre.

2.3 Per model: the pooled line averages over models that disagree

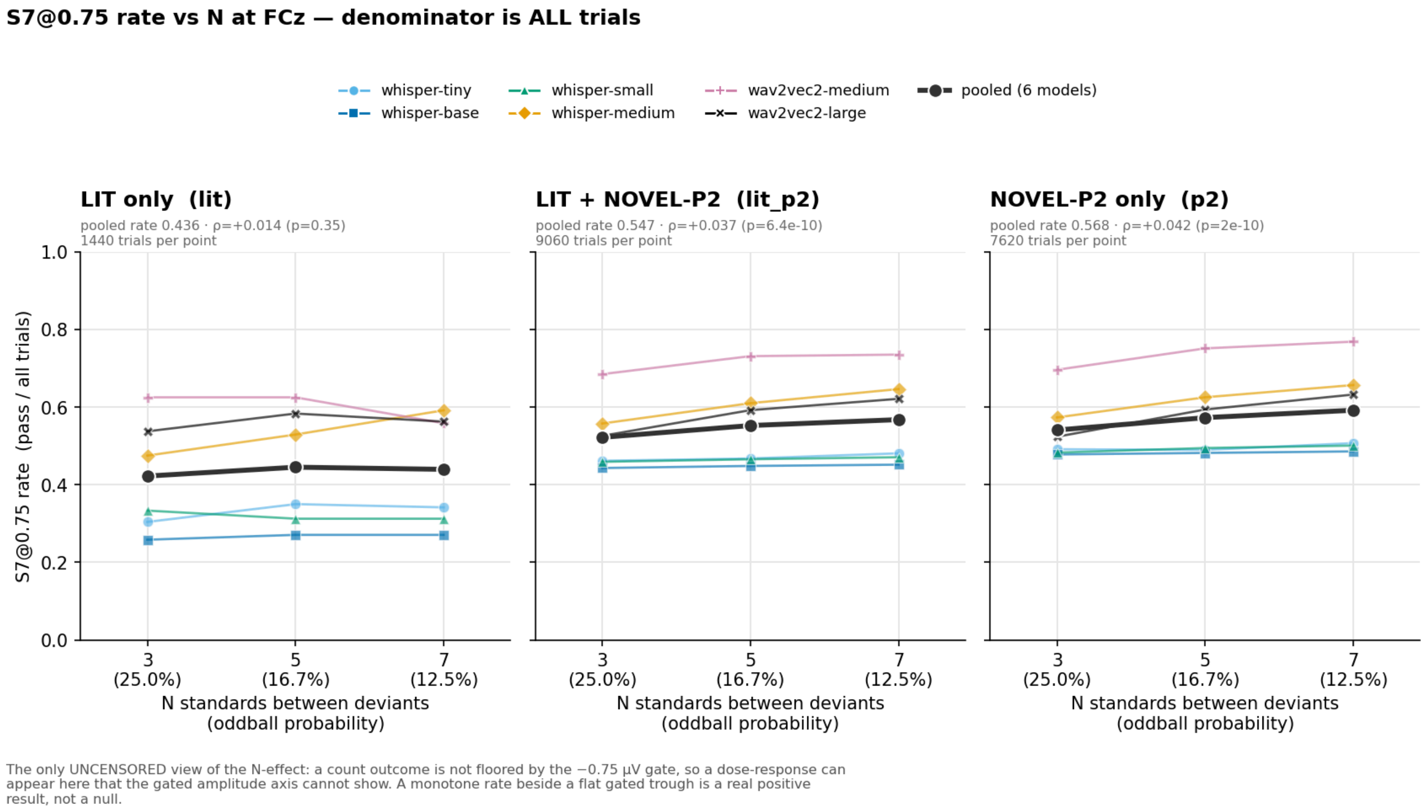


Median with 95% bootstrap CI of the S7@0.75-passing trials, over stimuli balanced across all three N levels. EACH PANEL HAS ITS OWN y-SCALE, shared with this figure's mean/median companion so the pair can be compared panel by panel — the models sit ~2 μV apart while each one's change across N is ~0.05–0.35 μV, so a shared axis would flatten every trend to a few percent of its height. Compare SHAPE across panels, not height; the pooled figure carries the cross-model amplitude comparison on a shared axis. ρ is Spearman at the (model, condition, N) cell level.

Within LIT, whisper-medium deepens in **77% of its stimuli** (median −0.483 μV, p = 6e-05), while wav2vec2-medium (35%), whisper-base (38%) and whisper-small (40%) run the *other* way. Only 3 of 6 models share a ρ sign between the two sources, so — contrary to the expectation that N is the identical manipulation in both — `lit_p2` **reads as a mixture for amplitude and is reported per source**, not as an n-boost.

Per-panel y-scales are used here deliberately: the models sit ~2 μV apart while each one's change across N is 0.05–0.35 μV, so a shared axis compresses every trend to a few percent of the axis height. Compare *shape* across panels, not height.

2.4 Pass rate: a steep, robust rise



set	N=3	N=5	N=7	Cochran–Armitage trend	rise
`lit`	0.422	0.445	0.440	z = +0.94, p = 0.35	+1.7 pts
`lit_p2`	0.522	0.552	0.568	z = +6.18, **p = 6e-10**	+4.6 pts
p2	0.541	0.573	0.592	z = +6.36, **p = 2e-10**	+5.1 pts

Carried by the larger models. Within NOVEL-P2 the trend is significant for whisper-medium (z = 4.33), wav2vec2-medium (z = 4.15) and wav2vec2-large (z = 5.55), all $p < 1e\text{-}4$, and absent in whisper-tiny, -base and -small (all $p > 0.35$). Suggestive for a scaling project — but six models across two architectures, matched on nothing but size, so a hypothesis rather than a result.

3. What the pass-rate effect actually is

The S7 pass rate is a product: **P(S7) = P(shape) × P(deep enough | shape)**. Decomposing it separates "the model produces an MMN-shaped response more often" from "the response it produces is bigger". Within NOVEL-P2:

	P(shape)	P(deep | shape)	P(S7)
N = 3	0.773	0.700	0.541
N = 5	0.780	0.734	0.573
N = 7	0.785	0.754	0.592
trend	+1.2 pts, ns	**z = +6.63, p = 3e-11**	z = +6.36

The entire N pass-rate effect is the depth term. The shape-pass rate barely moves. So the "rate" result is not a separate phenomenon from amplitude — it *is* the amplitude effect, measured by how often the trough clears a fixed threshold instead of by its conditional mean.

The same decomposition on deviance is more striking, because the two terms oppose:

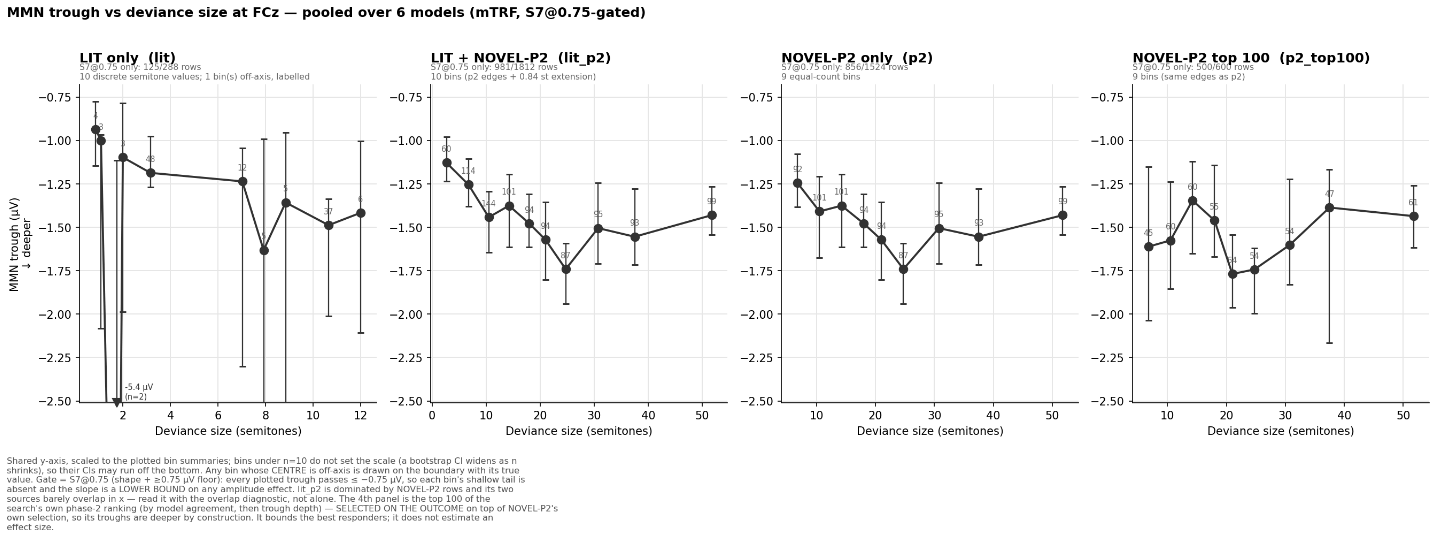
semitone tertile	P(shape)	P(deep | shape)	P(S7)
4.5 – 16.5 st	0.828	0.689	0.570
16.5 – 27.0 st	0.756	0.722	0.546
27.0 – 63.0 st	0.748	0.761	0.569
trend	**falls**	**ρ = +0.071, p = 0.015**	ρ = +0.005, ns

NOVEL-P2's flat S7 deviance rate is not a null — it is a rising amplitude effect cancelled by a falling shape-pass rate. Larger separations make the response bigger *and* make it less likely to have the classical dip-and-recover shape at all, which is consistent with the >2-octave conditions behaving like a different sound rather than a deviant (§4.3).

Why the gated amplitude looks flat. Conditioning on trough ≤ −0.75 μV truncates the distribution at a fixed point. Shifting a distribution deeper pushes more of it past the threshold — raising the *rate* — while the newly-admitted traces sit just past the floor and are therefore shallow, pulling the *surviving mean* back up. The two partly cancel, which is exactly what the S7-vs-S2 gap shows (−0.047 vs −0.085 μV). A flat gated trough beside a rising pass rate is the signature of a real but modest amplitude effect seen through a threshold, not of an effect on occurrence alone.

4. Deviance scaling

4.1 Pooled



Pooled Spearman ρ of trough vs semitones (negative = deeper with more deviance = MMN-like):

set	full range (S7)	≤ 24 st (S7)	full range (S2)	≤ 24 st (S2)
`lit`	−0.311 (p = 4e-04)	−0.311	−0.240 (p = 9e-05)	−0.240
`lit_p2`	−0.096 (p = 0.003)	−0.165 (p = 3e-05)	**−0.190 (p = 3e-13)**	−0.205 (p = 6e-11)
`p2`	−0.065 (p = 0.056)	−0.145 (p = 9e-04)	−0.093 (p = 0.001)	−0.102 (p = 0.006)

As with N, the ungated numbers are larger and more significant than the gated ones.

4.2 LIT's headline number does not survive decomposition

ρ = −0.311 is the largest single number in this deliverable. Split at the point where NOVEL-P2 begins:

LIT sub-range	ρ	p	n	median trough
0.84 – 3.16 st (low cluster)	−0.086	0.51	60	−1.13 µV
4.50 – 12 st (overlap)	−0.089	0.48	65	−1.36 µV
full range	**−0.311**	**4e-04**	125	−1.27 µV

Flat within each half. The whole correlation is the −1.13 → −1.36 µV *step* between two dense method clusters — 10 of 24 methods sit at 3.16 st and 5 at 10.65 st — which also differ in SOA (200–1000 ms) and tone duration (50–200 ms). It is a two-cluster contrast confounded with timing, structurally the same weakness as the combined-set slope, and should never be quoted alone.

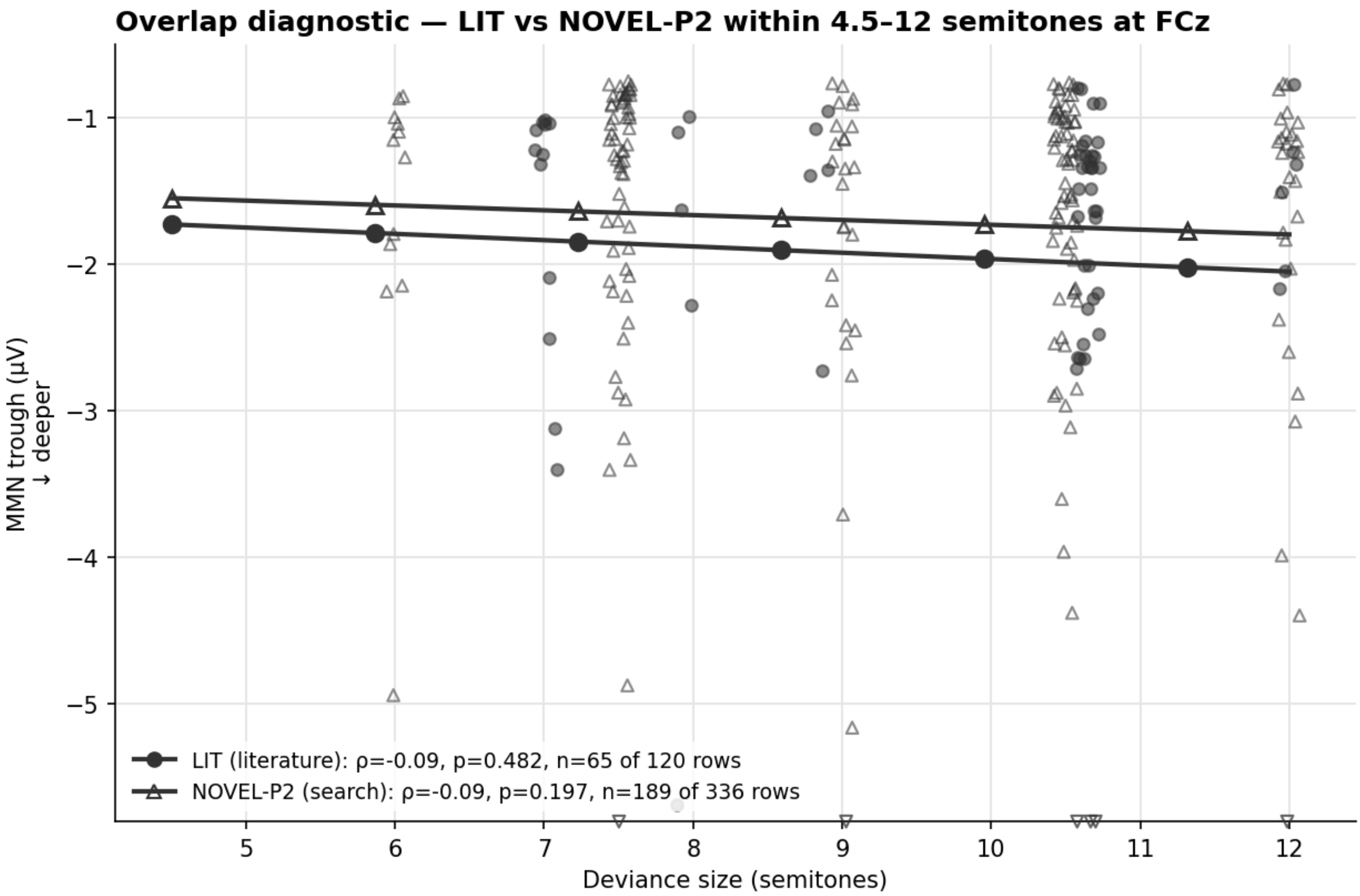
4.3 The effect is real below two octaves and reverses above

NOVEL-P2 is the only source with timing controlled, and it splits sharply at 24 st:

range	ρ	p	n	median
4.5 – 12 st	−0.094	0.20	189	−1.29 µV
12 – 24 st	−0.139	0.0095	347	−1.52 µV
> 24 st	**+0.131**	**0.015**	347	−1.55 µV
≤ 24 st	**−0.145**	**9e-04**	519	−1.46 µV

Saturation is itself MMN-like — human MMN amplitude saturates with large deviance. The *reversal* is more interesting: beyond two octaves the deviant appears to stop being processed as a deviation from the standard, which the falling shape-pass rate in §3 independently corroborates. It is also confounded with absolute frequency region, since large separations necessarily involve different parts of the spectrum. **ρ = −0.145 on the ≤ 24 st subset is the cleanest deviance estimate here**: timing fixed, 519 gated rows, and not a between-cluster or between-source contrast.

4.4 Is the combined set trustworthy?



The one region where both sources have conditions, so the two can be compared WITHOUT the range confound. Pooled over the 6 models, S7@0.75-gated, mTRF; lines are per-source OLS, labelled by the marker riding them (● LIT, △ NOVEL-P2). DESCENDING left-to-right = deeper with more deviance. ▽ 6 point(s) lie deeper than the axis and are drawn on the bottom edge; they still count in ρ and the fits.
If the two agree here, the lit_p2 combined slope is credible as a deviance effect; if they diverge, that slope is a source effect — the sources also differ in SOA, tone duration, deviant probability, and selection.

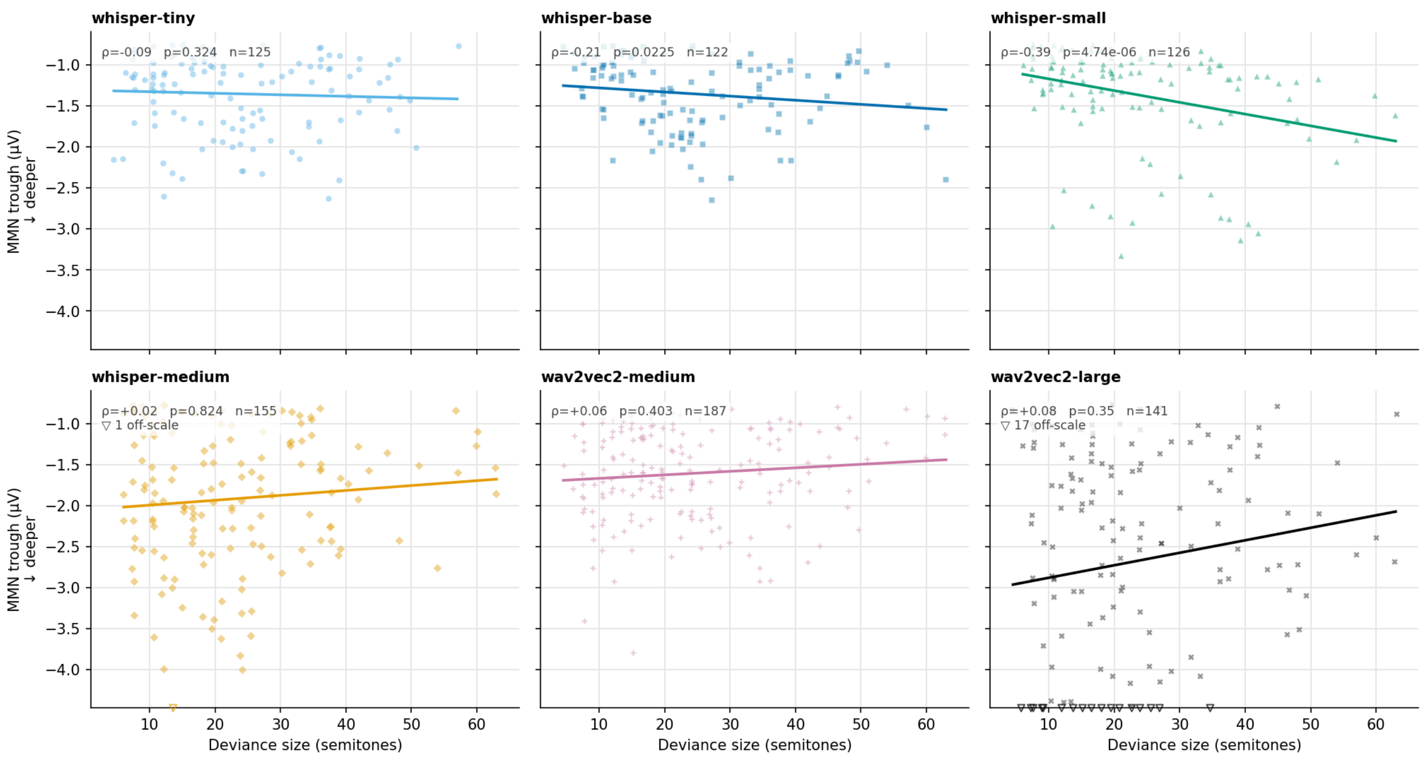
LIT spans 0.84–12 st and NOVEL-P2 spans 4.50–63, so a slope fitted across the union is substantially a **between-source contrast wearing a deviance label** — the sources also differ in SOA, duration, deviant probability and selection, and `lit_p2` is 87% NOVEL-P2 rows. Inside the 4.50–12 st window where both have conditions:

source	ρ	p	S7 rows
LIT	−0.089	0.482	65 / 120
NOVEL-P2	−0.094	0.197	189 / 336

Same sign, near-identical magnitude, near-identical median (−1.36 vs −1.29 μV) — but **both non-significant**. This is agreement on a *null*. It does the useful negative job of ruling out a systematic μV offset between sources driving the combined slope; it does **not** establish a deviance effect inside the overlap. The `lit_p2` ρ should never be quoted without this pair beside it.

4.5 Per model

MMN trough vs deviance at FCz — per model — NOVEL-P2 only (p2)



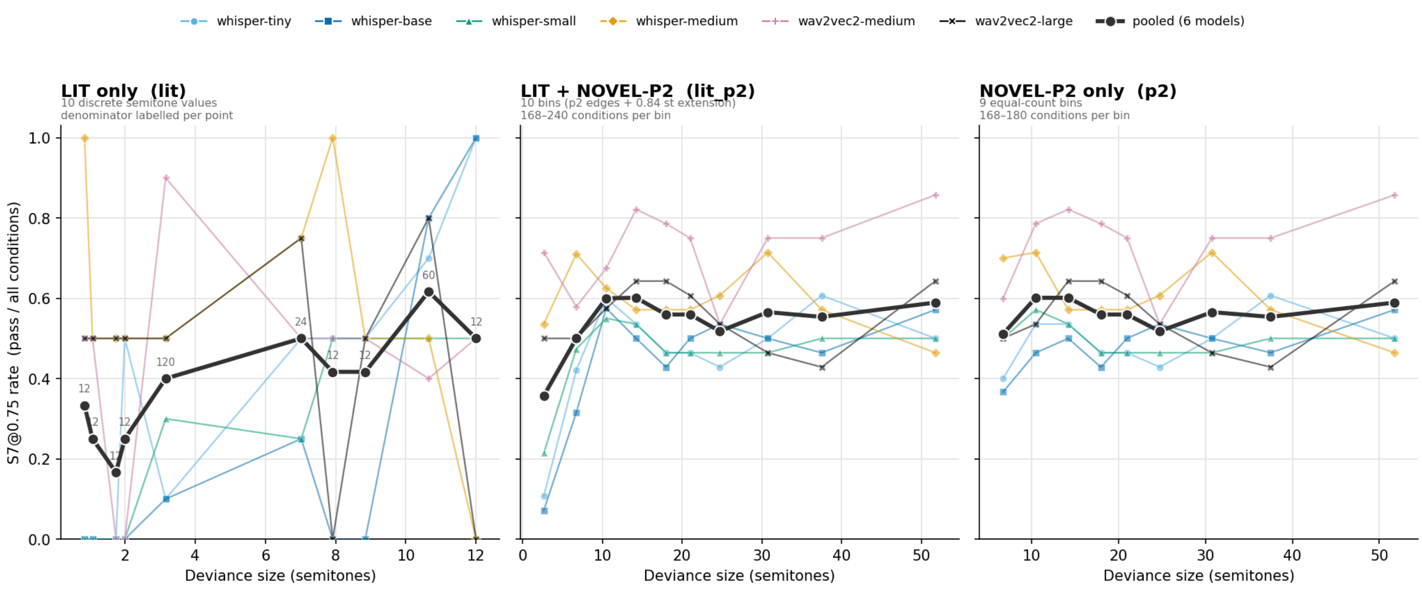
S7@0.75-gated troughs, mTRF. Shared y-axis across all six panels, clipped at the deepest 2% of the set's troughs (△ = points past it, drawn on the boundary and counted per panel). Line = OLS in linear semitones; ρ = Spearman (rank-based, computed on UNCLIPPED values).

Full-range ρ per model within `p2` (S7): whisper-small **–0.395** (p = 5e-06), whisper-base **–0.207** (p = 0.02), whisper-tiny –0.089 (ns), whisper-medium +0.018 (ns), wav2vec2-medium +0.062 (ns), wav2vec2-large +0.079 (ns). In LIT the strongest are whisper-small (–0.829) and whisper-tiny (–0.669).

So the deviance effect is carried by the **three smallest whisper models**, while whisper-medium and both wav2vec2 models are flat or marginally positive. That is close to the *opposite* of the N pass-rate split, where whisper-medium, wav2vec2-medium and wav2vec2-large carried the effect and the small whisper models were flat (§2.4). **The two dose-response axes are not picking out the same property of the models** — which is a caution against reading either as a general "this model is more MMN-like" ranking.

4.6 Pass rate

S7@0.75 rate vs deviance size at FCz — denominator is ALL conditions



The only UNCENSORED view in this deliverable: a count outcome is not floored by the −0.75 μV gate, so a dose-response can appear here that the gated amplitude axis cannot show. Bin edges match the amplitude figure within each set. LIT's per-model lines swing between 0 and 1 because most of its semitone values carry only 2 conditions — read the pooled line there.

set	first bin → last bin	trial-level ρ
`lit`	0.333 → 0.500	**+0.215** (p = 2e-04)
`lit_p2`	0.357 → 0.589	+0.069 (p = 0.003)
`p2`	0.511 → 0.589	+0.005 (p = 0.86)

Read this with §3: `p2`'s flat rate is two opposing effects cancelling, not an absence of one. The LIT rise is carried by the low-deviance end that only LIT covers.

---

5. Interpretation limits

5.1 N is confounded with oddball probability by construction. The generator sets rare-tone probability to 1/(N+1), so N = 3/5/7 means 25% / 16.7% / 12.5%. A deepening trough is genuinely MMN-like but cannot separate local spacing from global rarity; both are real mechanisms in humans. *(Note the generator's labels are inverted relative to the tones' roles — the frequent background tone is named "deviant" and the rare oddball "standard". The axis labels here are in role terms and are correct; see §18.7 of the repository overview.)*

5.2 NOVEL-P2 is selected on the outcome variable. Phase 2 admitted only pairs reaching ≥5/6 model agreement on S7 at FCz in phase 1 — essentially the quantity plotted here. It answers "*among stimuli that reliably evoke a model MMN, does amplitude scale?*", not "*does deviance drive amplitude?*". Its larger n does not make it the stronger claim.

5.3 NOVEL-P2's range is truncated from below by that selection. Of the full 903-pair grid, 0 of 42 pairs below 2 st and 0 of 41 between 2–4 st survived; the minimum selected separation is 4.50 st. The small-deviance end — where the effect should be weakest and the trend most diagnostic — is absent, and LIT is the only coverage there.

5.4 NOVEL-P2 trades a timing confound for a frequency-region one. Fixing SOA, duration and deviant probability is a genuine advantage over LIT. But `f_low` ranges 200–4,525 Hz and `f_high` up to 7,611 Hz, so large separations necessarily involve different absolute frequency regions where model and cochlear sensitivity differ.

5.5 LIT's coverage is lumpy and confounded with timing. See §4.2 — this is not a cosmetic caveat, it is the whole of LIT's −0.311.

5.6 Six models, not seven. whisper-large is excluded from every figure, table and statistic, for two independent reasons: its predicted μV run ~20–35× every other model (median S7-passing LIT trough at FCz −37.3 μV against a −1.08…−1.75 μV band for the other six), which would dominate any raw-μV mean and force a symlog axis; and it was never run in the novel search, so including it would make `lit` and `p2` incomparable. **It was not tested here — it did not quietly fail.**

5.7 The `p2_top100` panel is selected on the outcome twice. It is the first 100 rows of the search's own phase-2 ranking (by model agreement, then trough depth), on top of NOVEL-P2's own selection. Its S7 pass rate is 83% against p2's 56% and its troughs are deeper by construction (median −1.57 vs −1.49 μV). It bounds the best responders; it does not estimate an effect size, and no statistic in this memo is computed on it. Reassuringly, it reproduces p2's *shape* — the same non-monotone N pattern and the same dip-and-recovery in deviance — so the trends are not an artefact of averaging in weak responders.

5.8 Predicted μV are not literature EEG μV. Ridge shrinks the mTRF's predicted amplitude, and the 0.75 μV floor is calibrated to the models' own predicted-trough distribution, not to a clinical scale. Absolute depths (medians −1.27 to −1.57 μV) are **not** comparable to published MMN amplitudes. Only direction and monotonicity transfer.

5.9 Why this matters for clinical work. Essentially every clinical MMN finding — the schizophrenia literature especially — is an **amplitude reduction**. A readout whose amplitude tracks the driving manipulations only weakly is a correspondingly weak substrate for modelling that. §3 argues the underlying amplitude sensitivity is real but that our measurement compresses it; establishing how much is measurement and how much is the model is the highest-value follow-up.

6. Provenance and reproduction

artefact	path
per-trial scoring	`analyze_mmn_per_trial_n.py` → `mmn_per_trial_n_fcz.csv` (4,320 + 22,860 rows)
N figures + stats	`n_effect_plots.py` → 4 stats CSVs
deviance figures + stats	`deviance_scaling_s7gated.py` → 1 stats CSV
shared vocabulary	`mmn_dose_response_common.py`
figure-reading guide	`README.md`
cluster runbook	`handoff_per_trial_deviants_n_effect.md`

```
conda activate mbs-env
python aux/analysis_MMN_realness_checks/deviance_scaling_s7gated.py --gate s7   # and --gate s2
python aux/analysis_MMN_realness_checks/n_effect_plots.py           --gate s7   # and --gate s2
```

Output is byte-reproducible: fixed SVG hash salt, suppressed SVG timestamps, and a seeded bootstrap. Scoring primitives are imported from `analyze_mmn_criteria` / `analyze_mmn_criteria_s5_s6`, never reimplemented, with knobs identical to `analyze_mmn_s7_roi.py`. The driver patch that made the per-trial analysis possible, and the two cluster re-runs it required, are documented in §18 of `aux/sophies_repository_overview.md`.